

CSC 113 — Lab 6: List Exercises

Complete each function in lab6.py. Run the provided tests with: `python -m pytest test_lab6.py`

1. `average(nums)`: return the mean of a list of numbers, or 0 for an empty list.
2. `count_above(nums, cutoff)`: return how many values are strictly greater than cutoff.
3. `second_largest(nums)`: return the second-largest distinct value, or None if there isn't one.
4. `dedupe(items)`: return a new list with duplicates removed, keeping the original order.
5. `rotate(items, k)`: return the list rotated right by k positions (k may exceed the length).
6. Challenge: `running_totals(nums)` returns the list of cumulative sums.

Answers

1. `sum(nums) / len(nums)` if `nums` else 0
2. `sum(1 for n in nums if n > cutoff)`
3. Sort the set of values; return the second-to-last if there are at least two
4. Track seen values in a set and append unseen items to a result list
5. `k %= len(items)`; return `items[-k:] + items[:-k]` (handle the empty list)
6. Keep a running total and append it after adding each number